

A Report on

Movement Detection Using Transistor & PIR Module

for

**Self Learning Micro Project of Second Year, Semester III, Course :
Analog and Digital Circuit Design**

in

Electronics & Telecommunication Engineering

by

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SHAH AND ANCHOR KUTCHHI ENGINEERING COLLEGE

Academic Year: 2025-2026



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CERTIFICATE

This is to certify that the project entitled **Micro project Title** is a bonafide work of

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submitted to the Course In charge , Analog and Digital Systems Design , Shah and Anchor Kutchhi Engineering college , Mumbai in partial fulfillment of the requirement for the award of Self learning **Micro project , Second Year, (SY Sem-III)** in **Electronics & Telecommunication Engineering** during academic year **2025-26**

(Course In charge)



1 Title of Micro project : Movement Detection Using Transistor & PIR Module

2 Abstract

This project introduces an efficient and compact circuit capable of detecting human movement using a combination of a Passive Infrared (PIR) sensor and a transistor-based control system. The PIR module senses infrared variations caused by body heat, while the transistor amplifies this small analog signal to drive an output device. Unlike programmable systems, this circuit operates solely on hardware logic, offering a fast and low-cost solution. The simplicity and reliability of the setup make it suitable for smart electronic systems such as automated lighting, home security, and energy management applications.

3 Introduction

The integration of sensors and electronic switching devices is a key concept in modern circuit design. This project demonstrates how analog data from the environment can be translated into digital responses through basic electronic components.

By combining a PIR sensor with a transistor, the circuit detects motion and produces a visible and audible response, bridging the gap between analog sensing and digital actuation.

The project emphasizes how principles of *Analog and Digital Circuit Design (ADCD)* can be applied practically to create automation-oriented systems.

Objectives:

- To implement motion sensing using simple analog components.
- To demonstrate transistor operation in real-time switching.
- To achieve automatic control without microcontrollers.
- To explore practical automation concepts through circuit design.

4. Problem Definition

Conventional security or control systems rely on manual switching or programmed controllers, which increase cost and complexity.

This project addresses the need for an *independent, hardware-only motion detection system* that works efficiently with minimal components.

Input: Infrared radiation detected by the PIR sensor.

Process: Amplification and signal switching via transistor.

Output: LED and buzzer activation as motion indicators.

This circuit provides a low-maintenance and affordable automation solution for small-scale environments.

5. Literature Review (Optional)

Studies on motion sensing circuits show that combining analog sensors with transistor-based switching can replace costly microcontroller systems for basic detection needs.

PIR sensors such as HC-SR501 are proven to be reliable, energy-efficient, and ideal for short-range indoor use. Implementations found in DIY automation and research models confirm that transistor interfacing enhances detection precision while reducing false triggers.

6. System Design

The system is composed of three key parts — sensing, amplification, and output activation.

- **Sensing Unit:** PIR sensor identifies infrared variations caused by human movement.
- **Processing Unit:** Transistor BC547 amplifies the sensor signal and acts as an electronic switch.
- **Output Unit:** LED and buzzer provide real-time visual and sound alerts.

The system uses resistors to limit current and a regulated 5V supply for safe operation.

This design is scalable for advanced automation by adding relays or wireless modules.

7. Implementation

Hardware Components: PIR Sensor, BC547 Transistor, LED, Buzzer, 10k Ω & 220 Ω Resistors, Power Supply (5V), Connecting Wires, Breadboard or PCB.

Software Components: None – purely analog hardware setup.

Working Principle:

1. The PIR module detects movement by sensing changes in infrared energy.
2. When a person enters its range, the sensor's output voltage rises.
3. The transistor switches ON, completing the circuit for LED and buzzer.
4. Both indicators remain active until motion ceases.
5. Once the environment stabilizes, the sensor resets to standby mode.

This project verifies how a single transistor can act as both amplifier and switch, forming the core of a fully automated detection system.

8. Results & Observations

The circuit performed consistently within a range of 5–7 meters under standard indoor conditions. It responded quickly to human movement while maintaining low standby power consumption.

Condition	PIR output	Transistor State	LED & Buzzer
No Motion	Low	Off	Off
Motion Detected	High	On	On

9. Advantages and Limitations

Advantages:

- Entirely hardware-based – no code required

- Compact, durable, and cost-efficient
- Fast response time with minimal delay
- Demonstrates analog-to-digital conversion through switching

Limitations:

- Sensitive to ambient temperature and air movement
 - Works best for indoor conditions
 - Fixed detection range (limited to 5–6m)
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10. Applications

- Automatic motion-triggered lighting in hallways or classrooms
 - Motion alarms for doors, lockers, or safes
 - Smart energy-saving systems in homes/offices
 - Automation in labs or small industrial setups
 - Human detection for smart entry systems
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11. Conclusion

This project successfully demonstrates how analog and digital principles can be merged to create a simple yet effective motion detection circuit.

Using just a PIR sensor and a transistor, reliable motion-based automation was achieved without microcontrollers or programming.

The project highlights real-world implementation of ADCD concepts such as amplification, switching, and sensor interfacing.

Future enhancements may involve incorporating IoT modules, adaptive sensitivity controls, or mobile-based alerts for smarter operation.

12. References

- HC-SR501 PIR Sensor Datasheet
- “Electronic Devices and Circuit Theory” – Robert L. Boylestad & Louis Nashelsky
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